Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

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Abstract

Working note for the MicroBooNE search for Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions in the NuMI beam, $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$. The results are for the reverse-horn-current (RHC) exposure of 1.15×10^{21} POT; the forward-horn-current (FHC) exposure, whose wrong-sign $\bar{\nu}_\mu$ adds almost as much signal again, is to be included once its signal samples are complete. Two selections are maintained, with one signal definition: $\Lambda \to p\pi^-$ and the proton, pion and photon above this detector's reconstruction thresholds (KE 60, 30 and 30 MeV), 16.1 GENIE or 4.0 NuWro events in the fiducial volume. The PeLEE chain runs on standard PeLEE ntuples and ends in a cross-validated gradient BDT: $S = 0.57$ NuWro or 2.53 GENIE on $B = 128$. The Λ-style chain follows the MicroBooNE Λ CCQE analysis on HyperonProduction ntuples, fitting the Λ decay vertex, testing the connectedness of the raw wire activity and requiring a proton-like proton candidate: $S = 0.118$ NuWro or 0.475 GENIE on $B = 2.5$. Neither can observe Σ^0 at the predicted rate; the deliverable is an upper limit. The best median expected 90% CL limit is 8.7× the GENIE and 35× the NuWro rate, about 140 defined-signal events, from the Λ-style chain, which gives the better limit at every background systematic. The two generators differ by a factor of four in the predicted rate, almost entirely because NuWro includes hyperon final-state interactions. GENIE's $\Delta S = 1$ quasi-elastic cross section was found to be wrong in its hyperon-mass terms; every GENIE sample is corrected by event weights, and the GENIE signal samples are being regenerated with a fixed GENIE. The Λ-style chain's open items, data and dirt in its ntuple format, its beam-off estimate and its systematics, are the next pieces of work.

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1 Status

The analysis sets a limit on Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions,

$$\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\gamma, \qquad \Lambda \to p\,\pi^-, \tag{1}$$

so the reconstructed final state is a muon, a proton, a pion and a photon of about 75 MeV. All yields are scaled to the RHC data exposure, 1.1526×10^{21} POT.

Two selections exist, on two ntuple formats, and they are fully independent (Table 1). The PeLEE chain (`Sigma/pelee/SigmaSelection.C`, Sec. 3) is the analysis: it is the only one that can see data, EXT and dirt. The Λ-style chain (`SigmaLambdaSel.C`, `IslandProcess.C`, Sec. 4) needs the raw wire activity, which only the HyperonProduction ("hypana") format stores; that format exists for MC and EXT but not for data or dirt.

The main facts.

- **The result will be a limit.** The best median discovery significance at the predicted rate is about 0.3σ (GENIE) and 0.07σ (NuWro). The PeLEE ntuples carry the multisim weights the background covariance needs (`weightsGenie`, `weightsPPFX`, `weightsReint`).

Table 1: The two selections at their operating points, RHC. Signal yields carry every weight: the NuWro flux weight, PPFX on GENIE, the GENIE hyperon cross-section correction and the Σ^0 polarisation (Sec. 2). μ_{UL} is the median expected 90% CL limit on the rate in units of the model, at a 15% background systematic (Sec. 5).

	PeLEE BDT	Λ-style
ntuple format	PeLEE	HyperonProduction
data, EXT, dirt	all three	EXT only (factorised)
final discriminant	BDTG > 0.830	island test on ≥ 2 planes, proton PID
defined signal in the FV, NuWro / GENIE	3.98 / 16.05	
S, NuWro (with hyperon FSI)	0.572	0.118
S, GENIE (no hyperon FSI)	2.53	0.475
B (NuWro / GENIE model)	128.0 / 129.4	2.48 / 2.52
purity, NuWro / GENIE	0.44% / 1.9%	4.5% / 16%
μ_{UL} ($f = 15\%$), NuWro / GENIE	69 / 15.5 (cut re-optimised: 52 / 12.3)	36 / 8.8

- **The two signal models differ by a factor of four in rate** (49.3 against 12.6 in-FV Σ^0), almost all of it hyperon final-state interactions, which NuWro models and GENIE does not (Sec. 2.3). Their production rates agree to about 10% once GENIE's cross section is corrected. Every result is quoted under both.
- **GENIE's Σ^0 cross section was** $1.85\times$ **too high** and its Λ $0.8\times$ too low (Sec. 2.4, App. A). All GENIE samples carry a correcting weight; the GENIE signal samples are being regenerated with a fixed GENIE (Sec. 6).
- **One signal definition for both chains** requires the visible decay, $\Lambda \to p\pi^-$ with the decay products above threshold (Sec. 2.1). With it the Λ-style chain, which reconstructs the $p\pi^-$ pair, gives the better limit at every background systematic: at $f = 0$ by 2% (GENIE) and 12% (NuWro) over the PeLEE BDT retrained for this definition, at $f \geq 10\%$ by $1.4\times$. Its limit is Poisson-limited and almost flat in f.
- **FHC adds almost as much signal as RHC** ($0.90\times$ per POT, 47 expected GENIE Σ^0 in 1.10×10^{21} FHC POT against 54 in RHC); its background per POT will be higher (Sec. 2.5).

2 Signal, samples and signal models

2.1 Signal definition and fiducial volume

The signal is the same for both selections, both generators and every sample:

- $\bar{\nu}_\mu$ charged current with exactly one Σ^0 and no other meson or strange particle at the primary vertex;
- the true vertex in the fiducial volume, $3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding $675.1 < z < 775.1$ cm (looser than the CC1μ1π box: the signal is rare enough that acceptance is worth more than vertex quality);
- $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$;
- muon momentum above 0.15 GeV/c, proton kinetic energy above 60 MeV, π^- kinetic energy above 30 MeV, photon energy above 30 MeV.

Both selections also require the reconstructed vertex in the same volume. The proton, pion and photon thresholds are where this detector's reconstruction turns on: the momentum at which the efficiency to reconstruct the particle as the right type (track for p and π^-, shower for the photon) reaches half its plateau, measured on the signal samples and identical for both generators (App. B). Below them the selections have almost no efficiency: the thresholds remove a quarter of

the $\Lambda \to p\pi^-$ signal each (proton, pion) but only 1–4% of what the Λ-style chain selects. About 61% of in-FV Σ^0 events have $\Lambda \to p\pi^-$ and half of those pass the thresholds, so the defined signal is 32% of all in-FV Σ^0: 16.1 GENIE and 4.0 NuWro events in the RHC exposure.

The PeLEE truth record stops at the primary vertex, so the decay mode and the decay-product momenta come from the HyperonProduction truth of the same productions, matched per event (`ThreshStudy.C` → `sigdef_*.tsv`). Σ^0 events outside the definition (chiefly $\Lambda \to n\pi^0$ and soft decay products) are background, and since they are part of the Σ^0 production being tested, each model takes them from its own sample: GENIE events in the GENIE-model limits, NuWro events in the NuWro-model limits. The overlay's own Σ^0 events, whose decay the PeLEE record cannot see, stay out of the background.

2.2 Samples

Table 2: Samples, scaled to 1.1526×10^{21} POT. History and validation of each sample are in App. C.

format	sample	content	scale
PeLEE	Σ^0 signal, other hyperons	NuWro run3b overlay, 249 832 events, 1.3394×10^{24} POT	8.61×10^{-4}
	GENIE Σ^0, fake data	GENIE Σ^0 QE overlay, 250 555 events, 1.2300×10^{24} POT	9.37×10^{-4}
	ν background	RHC overlay Runs 1, 2, 3, 4a, 4b, 4c, 1.5463×10^{22} POT	0.0745
	dirt	RHC Runs 1, 2b, 3b, 4a, 4b, 1.7409×10^{21} POT	0.662
	beam-off (EXT)	run3b, 3.265×10^7 gates	0.750
hypana	Σ^0 signal (GENIE)	`sig_genie_qe.list`, the QE sample above, 2506 files	9.37×10^{-4}
	Σ^0 signal (NuWro)	the run3b production above	8.61×10^{-4}
	ν background	`bkg_overlay.list`, 14 164 files, 7.0488×10^{21} POT	0.1635
	beam-off (EXT)	16 570 events from 3 349 run3b files, 162 912 gates	150.25
	dirt, data	none	

Per-event weights:

- **NuMI flux.** NuWro's flux files came from the old NuMI flux (g4numi v6, bugged beamline geometry), while the overlay and data are on the current `g4_10_4` flux. Every NuWro event carries the official New/Old ratio weight in $(E_\nu, \theta_{\text{beam}})$, mean 0.677 and flat to 0.1% for the signal subset; 49.9k effective events of 58.5k. The RHC dirt is on the old flux too and takes the same map.
- **PPFX.** The overlay carries `ppfx_cv`; the GENIE QE sample, made from generator files, takes the overlay's ⟨`ppfx_cv`⟩ in true E_ν (`PpfxMap.C`), 0.934 on the signal. Hypana MC carries no PPFX weight; the Λ-style GENIE yield is multiplied by 0.934.
- **GENIE hyperon cross section and Σ^0 polarisation** (Sec. 2.4).

2.3 The two signal models

NuWro and GENIE predict very different Σ^0 yields (Table 3). The difference has two parts: the production cross section, where GENIE was wrong and is corrected, and hyperon final-state interactions, a genuine model difference.

Table 3: In-FV $\bar{\nu}_\mu$ CC Σ^0 per 1.1526×10^{21} POT. "Produced" counts the Σ^0 made at the vertex, "surviving" those that leave the nucleus as Σ^0, "defined" the signal of Sec. 2.1 (with the polarisation weight for GENIE).

	produced	surviving	defined
GENIE, Pais cross section (as generated)	83.3	83.3	–
GENIE, cross section corrected	55.8	55.8	16.1
NuWro (flux-weighted)	53.6	12.8	4.0

Hyperon final-state interactions. NuWro transports the hyperon through the nucleus; GENIE's INTRANUKE does not re-interact hyperons, so for GENIE produced equals surviving. In the NuWro sample only 25.3% of $\bar{\nu}_\mu$ CC events still carry a Σ^0, 27.9% carry a Λ ($\Sigma^0 N \to \Lambda N$), 10% a $\Sigma^\pm$, and 37.3% no strange particle at all (hyperons below the binding threshold are dropped; the build used, `nuwro_old` with `hyp_Eb` = 30 MeV, keeps 24% of the Σ^0 it produces). The corrected GENIE and NuWro agree on production to 0.96; the remaining factor 4.4 in surviving Σ^0 is FSI. The MicroBooNE Λ paper (arXiv:2212.07888) notes the same asymmetry: FSI *adds* Λ, so there NuWro sits above GENIE. A NuWro control run with ordinary CCQE instead of the hyperon channel reproduces GENIE's CCQE rate to 17%, so NuWro's flux and target normalisation is sound (App. C). The measured Λ cross section, $2.0^{+2.2}_{-1.7} \times 10^{-40}$ cm^2/Ar, brackets both generators. The factor of about two in Σ^0 reduction quoted for NuWro in the second-class-current study is not reproduced by this build; that is open.

2.4 GENIE's hyperon cross section and the correcting weights

With NuWro's cascade switched off, the generators agree on Λ production (NuWro/GENIE 1.07) but not on Σ^0 or Σ^- (0.59 each), with the same energy dependence in all three (Fig. 1, centre). Both use the same Cabibbo SU(3) form factors. An independent calculation from explicit Dirac traces, validated on Llewellyn Smith CCQE, gives $\Sigma^0/\Lambda = 0.19$ on a free proton, in agreement with NuWro (0.21 on argon), against GENIE's 0.45. GENIE's `PaisQELLambdaPXSec` reproduces the trace when the final baryon has the nucleon mass and diverges as the hyperon mass is raised: its hyperon-mass terms are wrong. Relative to the trace, GENIE's Λ is $\times 0.80$ and its Σ^0 and Σ^- $\times 1.85$. The derivation and the checks are in App. A.

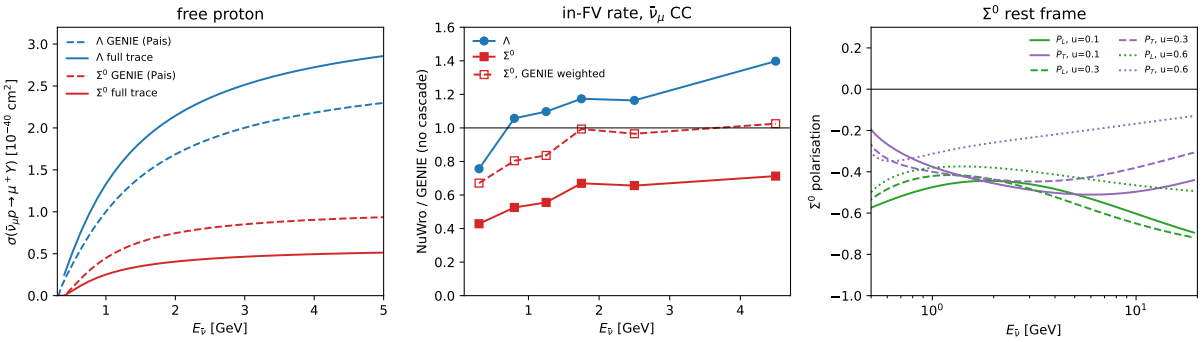


Figure 1: Left: $\bar{\nu}_\mu p \to \mu^+ Y$ on a free proton, GENIE's Pais splines (dashed) and the full trace with the same form factors (solid). Centre: NuWro/GENIE in-FV rate with the NuWro cascade off, flux-weighted, against true E_ν; open squares are GENIE Σ^0 with the cross-section weight. Right: Σ^0 polarisation in its rest frame against E_ν at three values of $u = (Q^2 - Q^2_{\min})/(Q^2_{\max} - Q^2_{\min})$.

Cross-section weight. $w(E,u) = [d\sigma/dQ^2]_{\text{trace}}/[d\sigma/dQ^2]_{\text{Pais}}$ on a free nucleon, with E the $\bar{\nu}_\mu$ energy in the struck-nucleon frame and $u = (Q^2 - Q^2_{\text{min}})/(Q^2_{\text{max}} - Q^2_{\text{min}})$ (`HypXsecWeight.h`, `hyp_xsec_weights.root`). For a two-body reaction E and Q^2 fix the final state up to a flat azimuth, so the weight corrects the kinematics as well as the rate; GENIE's phase space is populated everywhere except a corner holding $\leq 0.3\%$ of the corrected cross section (weight capped at 5). It costs no statistics. On the generator files:

in-FV per data POT	GENIE	GENIE weighted	NuWro (no cascade)	NuWro/GENIE weighted
Λ	209.5	263.0	223.7	0.85
Σ^0	81.7	54.4	47.9	0.88
Σ^-	194.9	129.3	115.0	0.89

The common 0.85–0.89 is a nuclear and flux difference. `SigmaSelection.C` applies the weight to every hyperon-QE event of every GENIE sample (overlay, dirt, detector variations, the QE sample), the struck nucleon taken at rest (1% effect); the hypana chains apply it by event identifier. NuWro is not weighted.

Σ^0 polarisation. The same traces give the Σ^0 polarisation, $P_L \approx P_T \approx -0.4$ to -0.5 (Fig. 1, right). Both generator chains hand Geant4 an unpolarised Σ^0, which decays isotropically. `PolWeights.C` weights each signal event by $w = 1 - \alpha(\vec{P}_\Sigma \cdot \hat{n})(\hat{n} \cdot \hat{q})$ ($\hat{n}$ the Λ direction in the Σ^0 frame, $\hat{q}$ the nucleon direction in the Λ frame, $\alpha = 0.747$), with the decay directions from the hypana ntuple of the same production. $\langle w \rangle = 0.997$, rms 0.15; the selected yields move by 0.2–1.4%.

What the weights cannot correct. The hyperon's nuclear potential and binding near threshold, which shift its kinetic energy, and hyperon FSI, which changes the topology; GENIE has neither. The polarisation weight is needed even with a fixed GENIE.

2.5 FHC

The wrong-sign $\bar{\nu}_\mu$ in FHC gives 0.90× the RHC Σ^0 yield per POT (GENIE, corrected: 49.2 against 54.4 per 1.1526×10^{21} POT) with a similar spectrum ($\langle E_\nu \rangle$ 1.56 against 1.48 GeV). With the 1.1048×10^{21} POT of all FHC runs, including Run 5, that is 47 expected in-FV Σ^0, 87% of RHC. The selection does not use the muon charge, so FHC's ν_μ-dominated background per POT will be larger. The PeLEE FHC overlay (Runs 1, 2, 4c, 4d, 5) and beam-off samples exist; the FHC GENIE signal sample is being regenerated (Sec. 6). FHC enters as a second bin of a combined limit.

3 The PeLEE selection

3.1 Candidates and cut chain

- **Muon**: longest track with `trk_score` > 0.8, start within 20 cm of the vertex, length > 10 cm, LLR PID > 0.7.
- **Proton and pion**: each non-muon track is scored for both roles and the best ordered pair is taken (`PidAssign.h`); pion-candidate purity 42.5% against a ceiling of 51.4% (the fraction of selected signal with a reconstructed true π^-). The candidates are formed when the tracks exist but are not required: requiring them removes more signal than background. Events without a pair carry sentinels and the BDT input `HasPair`.
- **Photon**: the non-track PFPs closest to the vertex; the closest is the Σ^0 decay photon, 70 MeV on average in signal against 74.5 expected.

Stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\mathrm{shr}} < 4$), `MuonCandidate`, `NTracksCut` ($n_{\mathrm{trk}} < 5$), `BDTG`.

3.2 BDT

Gradient BDT (600 trees, depth 3) over 57 inputs: candidate PID and geometry, Λ and Σ^0 masses, Λ and photon vertex pointing, Λ decay distance, a π^0 di-photon mass, the Molière angle and the second-shower-veto block of the single-photon analyses (App. E). Two-fold cross-validation by event parity: each fold is applied only to events it did not train on, so every BDT number is out of sample. Each fold trains on 95% of its events; the signal class is the NuWro and GENIE QE signal of the definition of Sec. 2.1 (GENIE scaled to the NuWro summed weight); the background class holds the overlay, dirt, EXT, NuWro's non-signal hyperons and the Σ^0 of both generators outside the definition, so the BDT learns to prefer a real $p\pi^-$ pair; every input is mapped into its bulk range (`BDTClamp.h`), because the ntuples carry $\pm$FLT_MAX, 10^5 and INT_MAX sentinels that defeat TMVA's evenly spaced cut grid. Retrained for the definition, it halves the out-of-definition Σ^0 it accepts and lowers the background by a third at the previous cut. The development measurements are in App. D.

3.3 Results

Table 4: PeLEE chain, signal as defined in Sec. 2.1, all weights. Raw signal and S are NuWro; GENIE in Table 5. B = NuWro non-signal hyperons (here including NuWro Σ^0 outside the definition, the NuWro model) + overlay + dirt + EXT.

stage	raw signal	S	other hyperons	overlay	dirt	EXT	B
AllEvents	6930	3.98	49.91	303821	282305	2406133	2992330
FiducialV	5762	3.33	38.44	90642	22496	148091	261268
CosmicVeto	5189	2.98	32.54	62204	7720	22928	92885
NShowers	3911	2.25	16.78	28995	2906	11942	43860
MuonCandidate	3621	2.06	13.73	20348	1777	6844	28983
NTracksCut	3497	2.00	13.63	18053	1756	6730	26553
BDTG > 0.830	**1000**	**0.572**	**0.69**	**127.1**	**0.22**	**0.00**	**128.0**

Efficiency 14.4% (NuWro, weighted), purity 0.44%. At the cut EXT has no raw event left (it is given the error of one event of its own weight). The counting figure of merit peaks at 0.95 on a background of a few raw-weighted events, which is not a usable point; 0.830 minimises the expected limit at $f = 0$–5%, and at larger f the limit prefers 0.89–0.95, where the background rests on few raw events (Sec. 5). Figure 2 (previous training) shows the score shape. The surviving background is dominated by resonant and DIS events with a π^0 (App. E).

3.4 GENIE signal

The GENIE QE sample and the overlay's own GENIE Σ^0 events run through the full selection and are counted apart from every sample. The BDT trains on the QE sample's signal with the same parity split, so its yield is out of sample; the overlay's Σ^0 are never trained on. The two agree to 1.5% at every cut.

On the base definition the QE sample and the overlay agree to 1–2% at every cut. The efficiencies for the defined signal are 15.8% (GENIE) and 14.4% (NuWro). Σ^0 outside the definition passing the cut: 1.92 (GENIE) and 0.45 (NuWro), background in the corresponding model.

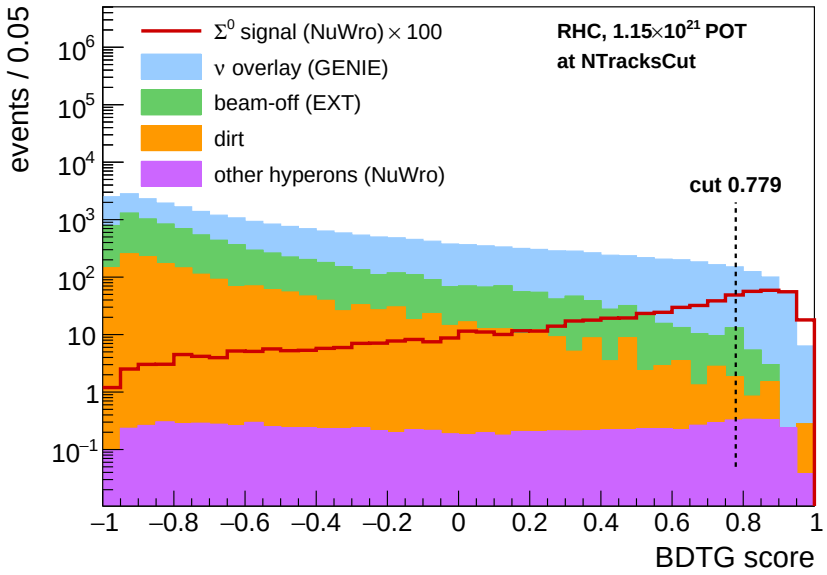


Figure 2: Cross-validated BDTG score at `NTracksCut`, background stacked by sample and the NuWro signal scaled by 100, for the training before the signal definition was restricted (cut 0.779, dashed).

Table 5: GENIE signal through the PeLEE chain, with the cross-section weight (both) and the polarisation weight (QE sample). The QE sample is the defined signal; the overlay's own Σ^0 cannot be restricted (no decay truth), and the ratio column compares the QE sample with it on the base definition.

	QE sample			overlay Σ^0		
stage	raw	S_{GENIE}	eff. [%]	raw	S	GENIE/NuWro
AllEvents (true FV)	27385	16.06	100	1036	48.51	4.04
MuonCandidate	14303	8.38	52.2	553	25.80	4.06
NTracksCut	13915	8.15	50.7	544	25.41	4.08
BDTG	4326	2.53 ± 0.04	15.8	95	4.50 ± 0.49	4.42

3.5 Detector variations

The Run 3 RHC detector-variation set re-simulates the same events under nine variations; `DetVar.C` compares each with the CV on shared events (old-flux samples, so only ratios are used).

variation	S var/CV	B var/CV
LY attenuation	1.000 ± 0.000	1.005 ± 0.005
LY down	0.975 ± 0.025	0.991 ± 0.006
LY Rayleigh	0.975 ± 0.039	0.984 ± 0.009
recombination	1.121 ± 0.072	0.926 ± 0.030
SCE	0.900 ± 0.108	0.972 ± 0.035
WireMod θ_{XZ}	0.992 ± 0.078	0.970 ± 0.025
WireMod θ_{YZ}	0.906 ± 0.087	0.976 ± 0.027
WireMod X	0.956 ± 0.078	0.972 ± 0.027
WireMod YZ	0.945 ± 0.081	0.985 ± 0.025
quadrature (MC noise alone)	19.9% (21.3%)	9.6% (7.0%)

At 0.779 with the training before the definition was restricted (to be redone at the current cut and training), with all weights. The signal is not resolved (its quadrature sum equals the MC noise). The background is: recombination lowers it by $7.4 \pm 3.0\%$ (2.5σ), and the sum exceeds the noise by about 6.6% in quadrature. The CV keeps 51 raw signal and 1 174 raw background events, so the background detector systematic is about 7–10% and the signal one is not measured.

4 The Λ-style selection

The MicroBooNE Λ CCQE analysis (Thorpe & Nowak, Hyperon Technote v2) selects $\bar{\nu}_\mu$ CC Λ with 91% purity, where the earlier hypana BDT chain for Σ^0 reached 2.8% at the same efficiency. This chain follows it:

- **Every ordered track pair is scored**; the pair selector's only topological input is the separation of the two track starts, which does not penalise a Λ that flew. Pair precuts are a preference, not a veto.
- **The decay vertex is fitted** (V fit), giving a decay length and the pointing angle α between the pair momentum and the vertex-to-decay direction.
- **The island test** (Fig. 3): on each plane a wire × time grid of the raw activity is clustered from seeds at the muon, proton and pion starts, and the proton and pion clusters must merge while the muon stays separate. Connectivity 2 wires × 20 ticks passes 80% of signal with 2–3 cm Λ flight. It needs the raw wire windows, hence hypana.

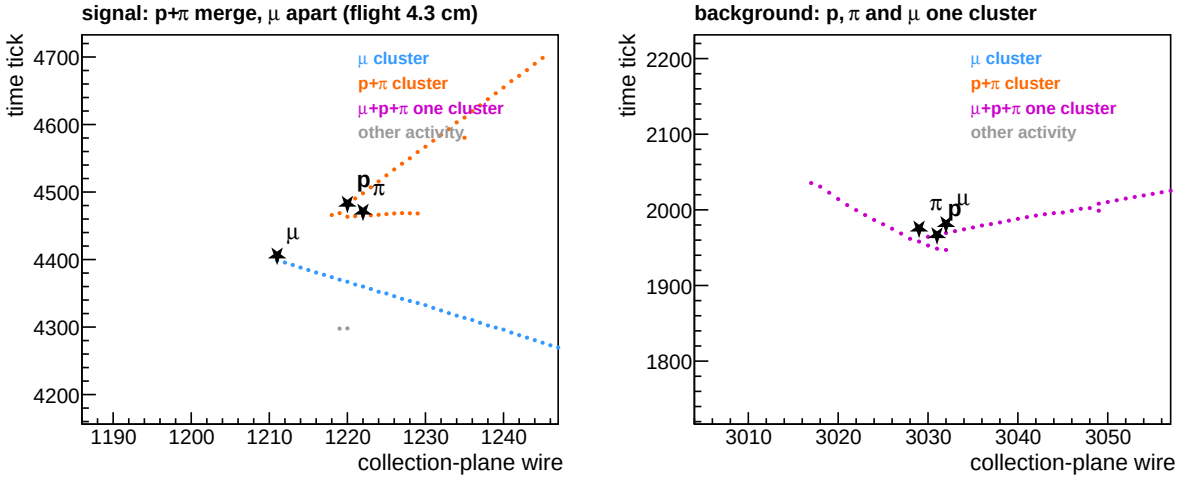


Figure 3: The island test on the collection plane. Left: a GENIE signal event with 4.3 cm true Λ flight; the proton and pion seeds share a cluster (orange) and the muon has its own (blue). Right: a background event at the same stage, whose proton, pion and muon seeds land in one cluster.

4.1 Operating point and results

Preselection as the PeLEE chain (flash-matched neutrino slice, reco vertex in the FV, topological score > 0.20, $0 < n_{\mathrm{sh}} < 4$, $n_{\mathrm{tr}} < 5$, a muon candidate), then Λ mass within ±20 MeV, decay length > 0.5 cm, $\cos\alpha > 0.95$, photon start > 2 cm from the vertex, island test passed on ≥ 2 planes, and a proton-like proton candidate (`llrP` < 0, Sec. 4.4).

The preselection keeps 51.7% of the defined signal (GENIE; NuWro 51.5%). By step: neutrino slice −14.7%, reco FV −2.6%, topological score −6.2%, shower and track counts −27.1% (of the signal reaching that step; mostly events with no shower-like PFP, 19% of all defined signal, where

the Σ^0 photon was missed or reconstructed as a track), muon candidate −9.0% (`PreselFlow.C`). The two generators agree at every step to 1%.

Table 6: Λ-style cut flow, signal as defined in Sec. 2.1. S is the GENIE Σ^0 QE sample with the cross-section, polarisation and PPFX weights, S_N the NuWro sample. "Out" is Σ^0 outside the definition (GENIE / NuWro), background in its own model. B is the hypana overlay, with the cross-section weight at the last stages. Errors are MC statistics.

stage	raw S	S (GENIE)	S_N	out (G / N)	raw bkg	B
MuonCandidate	14154	8.24	2.04	17.5 / 4.48	123000	20113
PairCandidate	6875	4.02	1.00	3.49 / 0.99	51238	8378
Λ mass ±20 MeV	4461	2.59	0.625	1.71 / 0.50	16391	2680
pointing ($\cos\alpha$, decay length)	1780	1.04	0.240	0.16 / 0.056	706	115.4
photon gap > 2 cm	1702	1.00	0.230	0.15 / 0.052	549	89.8
island, ≥ 2 planes	875	0.514	0.131	0.047 / 0.010	23	3.92 ± 0.82
proton PID, `llrP` < 0	**810**	$\mathbf{0.475 \pm 0.017}$	**0.118**	**0.041 / 0.008**	**12**	$\mathbf{2.12 \pm 0.62}$

With the NuWro signal $S = 0.118 \pm 0.009$ (207 raw). NuWro's own non-signal events (Σ^0 that FSI turned into something else) add 0.020, EXT 0.34 ± 0.15 (Sec. 4.2) and the out-of-definition Σ^0 0.008 (NuWro) or 0.041 (GENIE), so $B = 2.49$ (NuWro model) or 2.52 (GENIE model). The pair and pointing steps separate defined from undefined Σ^0: 2:1 against the signal at the muon stage, 1:12 at the end. The efficiency for the defined signal is 3.0% for both generators.

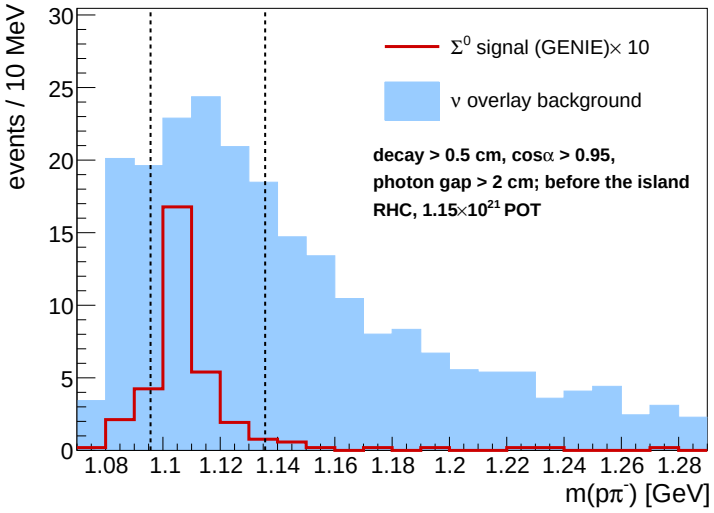


Figure 4: $p\pi^-$ mass after the pointing cut and the photon gap, before the mass window (dashed) and the island test. The pion momentum is taken from range under the muon hypothesis, 20% low for a pion but with a narrower peak than any correction. The background is largely genuine $p\pi$ from Δ decay.

4.2 Beam-off background

The run3b `ext_unbiased` files were taken through reco2 and HyperonProduction: 16 570 events, 0.67% of the data's cosmic statistics, too little to follow to the operating point (none of 76 MuonCandidate events reaches the pointing cut). `ExtFactor.C` factorises: EXT at MuonCandidate times the pass fraction of the overlay's cosmic-dominated candidates as far as the pointing cut, and the neutrino-background rate (3.3%) for the island and gap steps. EXT is 0.62 ± 0.27 events before the PID cut and, scaled by the MC background's PID pass fraction (0.54), 0.34 ± 0.15 **at**

the operating point. Without the neutrino-rate assumption for the island step the bound is < 6.2 at 90% CL; that, and the PID scaling, are the open part of this estimate. The real cosmic data agree with the factorisation wherever they still have events (8 of 76 EXT candidates have a pair, against 0.135 predicted from the cosmic MC and 0.42 from the neutrino rate). Details are in App. G.

4.3 What the remaining background is

The 23 overlay events at the island stage were read back from the hypana files with their full truth (`BkgDump.C`: neutrino, interaction, primaries, the origin of every PFP). Table 7 is the background at the operating point.

Table 7: Λ-style background at the operating point by event truth, NuWro model. The real Λ events carry GENIE's corrected Λ cross section (weight ≈ 1.25 for QE).

component	raw	B	share
real $\Lambda \to p\pi^-$, $\bar{\nu}_\mu$ CC QE ($\bar{\nu}_\mu p \to \mu^+\Lambda$)	4	0.81	33%
real $\Lambda \to p\pi^-$, associated production (ν_μ CC DIS $\Lambda K^0\pi^0$, ν NC RES ΛK^+)	2	0.33	13%
Σ^+ (ν_μ CC DIS $\Sigma^+ K^0$; muon candidate the π^+ of $\Sigma^+ \to n\pi^+$)	1	0.16	7%
no strange particle: pp pair (pion candidate a proton)	2	0.33	13%
$\pi^+\pi^-$ pair (NC DIS)	1	0.16	7%
secondaries from n/π^+ reinteraction	1	0.16	7%
ν_e NC, true vertex outside the FV	1	0.16	7%
EXT (factorised, Sec. 4.2)	–	0.34	14%
NuWro non-signal hyperons; Σ^0 outside the definition	–	0.03	1%
total	12	2.49	

- **Real Λ is 46% of the background.** The $\bar{\nu}_\mu p \to \mu^+\Lambda$ events are signal-like in everything but the photon: their "photon" is two δ-ray electrons on the muon, the Michel positron of the μ^+ (111 cm out) and a cosmic (452 cm). The photon handles were measured (Sec. 4.4) and do not improve the limit. In the associated-production events the photon is real, from a π^0 or from secondaries.
- **The muon candidate is not a μ^+ in 7 of 12 events** (4 μ^-, 3 π^+), the wrong-sign and NC part. Without a charge measurement the Michel tag is the only handle; it was marginal on the full background (App. G), and here 5 of the 12 have a contained muon with a PFP at its end.
- **The two pp pairs** have pion-candidate `llrPi` of 0.04 and −0.14, but 23% of the signal has `llrPi` < 0; a pion PID cut costs more than they weigh.
- **Σ^0 outside the definition** at the operating point: 62% fail only $\mathrm{KE}_\pi > 30$ MeV, 16% $\mathrm{KE}_p > 60$ MeV, 15% are $\Lambda \to n\pi^0$; 0.008 (NuWro) or 0.041 (GENIE) events.
- **The PID cut removes** 11 events: 4 real Λ with the proton and pion candidates swapped by the pair assignment, the rest with an e, γ or π as the proton candidate.

With 12 raw MC events (±29%) and a factorised EXT, more background MC and a data-driven cosmic estimate are worth more than another cut.

4.4 Selection optimisation

The figure of merit is the expected limit; at $B \approx 4.6$ it is set by Poisson counting, so signal efficiency matters as much as rejection. The background rests on 23 raw events, so every candidate cut is checked at the photon-gap stage (549 raw) and by parity cross-validation (`IslandOptimise.py`).

- **Proton-candidate PID (`llrP` < 0): adopted.** It keeps 91.5% of signal for both generators and removes about half the background. The expected limit improves by 12–13% on the median and on the mean over outcomes, and on both parity halves; its threshold is the natural midpoint, not tuned.

- **After it, the real Λ is 38% of the background.** Its pair is genuine, so only the photon separates it. A maximum photon distance removes it but loses the 16% of signal photons that convert beyond 55 cm, and does not improve the limit. Shower shape (hits per plane, length, opening angle, Molière angle, dE/dx) does not separate the Michel and δ-ray electrons from the soft Σ^0 photon. Photon pointing to the vertex rejects the real Λ but the 45 MeV photon is too poorly reconstructed for a useful cut. The Σ^0 mass window costs more signal than it removes. **A muon-track photon veto** (reject the event if its photon starts within 5 cm of the muon axis and more than 20 cm along it, up to 10 cm past its end: δ rays and Michel electrons) removes 2 of the 4 real-Λ events for 6% of signal, but leaves the mean expected limit unchanged (9.32 against 9.38 GENIE); it is not used. Without the 20 cm the veto would catch 23% of true Σ^0 photons, those converting near the vertex in the muon's direction (App. G).

- **The last three cuts and the shower ceiling, re-scanned with the full signal definition** (photon gap, island planes, `llrP`, $n_{\mathrm{sh}} \leq 1, 2, 3$; `LastCutsScan.py`): the operating point is the best of the 672 combinations under both models, and on the mean expected limit it sits on a plateau (gap 2–3 cm, $n_{\mathrm{sh}} \leq 2$ or 3, all within 1%). Cross-validated by parity, a choice made on one half is worse than the operating point on the other, so the scan resolves nothing finer than about 10%; the final background is 12 raw events and each step moves 1–4 of them (App. G.1).

- **Loosening any single cut** (mass window, pointing, decay length, photon gap, island planes, PID threshold) recovers signal only with proportionally more background and does not improve the limit. A full scan finds 20% on the full sample but about 6% out of sample; the background statistics do not support a finer optimisation.

- **Events without a reconstructed shower** (the soft Σ^0 photon missed, 20% of the signal at that stage) were taken through the chain as a second bin: $S = 0.213$ GENIE on $B = 5.64$ (EXT 1.53 ± 0.88 measured with the hypana EXT sample), purity 3.7%. Merged with the main bin they worsen the limit; as a separate bin they would gain 7–8%, which is not worth a second category. They are not used (App. G).

5 Expected sensitivity

5.1 Method

`Sensitivity.C` gives the expected 90% CL upper limit on μ, the Σ^0 rate in units of a generator's prediction, for both chains under both generators. Each chain is a single-bin counting experiment, $n \sim \mathrm{Pois}(\mu S e + b)$:

- b is lognormal with mean B and fractional width $\sqrt{r_{\mathrm{MC}}^2 + f^2}$, r_{MC} the MC statistics of B and f the background systematic, which is scanned;

- e is lognormal with mean 1 and width $\sqrt{g^2 + r_S^2}$, $g = 20\%$ a placeholder for flux, efficiency and POT;

- Bayesian limit, flat prior in $\mu \geq 0$, b and e marginalised (hybrid CL_s is 4% tighter; both reproduce the PDG table for known b);

- expected limit at the median n of Pois(B), band from the 16% and 84% quantiles; Z_A from Cowan et al. (arXiv:1007.1727).

$\mu_{\mathrm{UL}} = n_{\mathrm{UL}}(B, \sigma_b)/S$: the generator only rescales the limit, and the cut that minimises it does not depend on the signal normalisation.

5.2 Results

Table 8: Median expected 90% CL limit on μ, the rate of the signal of Sec. 2.1 in units of each model's prediction, RHC, $g = 20\%$, with all weights and each model's own out-of-definition Σ^0 in the background. "Re-optimised" scans the BDT cut from 0.50 to 0.95 separately for each f and model (cut in brackets, NuWro / GENIE).

f	PeLEE, BDTG > 0.830		PeLEE, re-optimised			Λ-style	
	NuWro	GENIE	NuWro	GENIE	cut	NuWro	GENIE
0%	39.7	8.87	39.7	8.87	0.83 / 0.83	**35.3**	**8.69**
5%	44.4	9.94	44.4	9.94	0.83 / 0.83	**35.4**	**8.70**
10%	55.3	12.45	50.8	11.92	0.89 / 0.885	**35.5**	**8.72**
15%	68.6	15.47	51.6	12.29	0.91 / 0.95	**35.6**	**8.76**
20%	82.2	18.58	53.6	12.30	0.93 / 0.95	**35.8**	**8.81**
30%	107.9	24.45	55.8	12.32	0.95 / 0.95	**36.3**	**8.94**

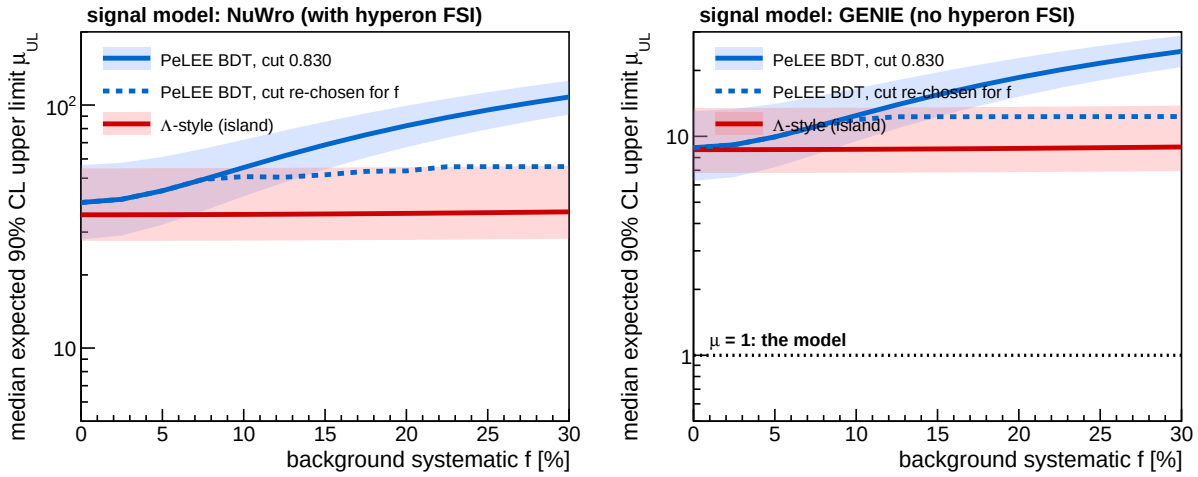


Figure 5: Median expected 90% CL limit on μ against the background systematic, under NuWro (left) and GENIE (right). Bands are the 16–84% range; the dotted line on the right is the GENIE prediction itself.

1. **The best expected limit**, 8.7× GENIE and 35× NuWro, is about 140 defined-signal events in 1.15×10^{21} POT, against 16 predicted by GENIE and 4.0 by NuWro.

2. **The Λ-style chain gives the better limit at every** f, under both models: at $f = 0$ by 2% (GENIE) and 12% (NuWro) over the PeLEE BDT retrained for this definition, at $f \geq 10\%$ by about 1.4×. It requires the reconstructed $p\pi^-$ pair, so its efficiency for the defined signal (3.0%) is three times its efficiency for all Σ^0; the PeLEE BDT, retrained, still accepts out-of-definition Σ^0 (1.9 GENIE events at the cut). At $B \approx 2.5$ the Λ-style limit is Poisson-limited: removing all background uncertainty moves it by 2%.

3. **The background systematic drives the PeLEE result.** Its limit grows 1.3–1.4× from $f = 0$ to 15% with the cut re-optimised and 1.7× at 0.830. The tightest re-optimised cuts (0.91–0.95) rest on few background MC events and are fragile.

4. **Nothing missing from the Λ-style chain changes its limit much** ($f = 15\%$, NuWro): without EXT 36.9 against 35.6; with dirt at its bound (8.7% of B_{MC}) 35.1; with $g = 0$ 32.6; with B known exactly 34.8. The median n is an integer (2 here), so small changes in B move the median limit in steps, either way. Since it carries the result, hypana data and dirt are needed rather than optional.

6 Open items and plan

1. **Samples (running or next).**

   - Fixed-GENIE (`genie::HypQELTracePXSec`, App. A.1) signal: FHC first, then RHC, at the lowest batch priority; then overlay, reco2, PeLEE and hypana ntuples. With them the cross-section weight is dropped for the signal; the polarisation weight stays, and the overlay and dirt keep the cross-section weight.
   - NuWro FHC signal overlay (generation complete; overlay to start).
   - NuWro hyperon background samples, per beam: QE Λ and Σ^- (S1) and associated strangeness, filtered to events with a hyperon (S2), so that the NuWro model has its own hyperon background (here 46% of the Λ-style background is real Λ from GENIE). Generator-level tests first.
   - A larger Run 1 EXT prestage (needed for FHC; done from a FNAL gpvm).

2. **Λ-style chain as the primary analysis.**

   - Data, dirt and a larger EXT sample through reco2 and HyperonProduction; replace the factorised EXT (0.34 ± 0.15, 14% of B) by a direct or data-driven estimate.
   - More background MC: B rests on 12 raw events (±29%), the leading uncertainty after Poisson.
   - Systematics: flux, cross section (with the hyperon cross section and FSI), reinteraction, and a hypana detector-variation set; the island test's data/MC agreement and the sidebands (Λ mass, island fail, PID fail).
   - The island retune on the QE sample, with the preselection kept as it is.

3. **FHC** (Sec. 2.5): both chains on FHC, the FHC signal efficiency from the regenerated samples, a combined two-bin limit with correlated signal normalisation.

4. **PeLEE chain as the cross-check.** The PPFX, GENIE and reinteraction multisims at the single bin; detector variations re-measured at the current cut and training; a data constraint from the low-score region (the π^0 normalisation).

5. **Λ cross-check** against the published Λ CCQE result with the corrected GENIE (App. G); report the GENIE hyperon cross-section problem to the GENIE authors (it moves GENIE's Λ by ×1.26).

6. **Signal model.** Hyperon FSI is the factor four between the generators; the NuWro build does not reproduce the published FSI reduction. The limit is quoted under both.

7. **Closure and unblinding.** Fake-data closure (Poisson throws, GENIE as the alternative model); then the observed limit from RHC (and FHC) beam-on data and the absolute cross-section limit from the integrated $\bar{\nu}_\mu$ flux and the argon count.

A GENIE's hyperon cross section in detail

Generators with the cascade off. NuWro was rerun with no cascade and no jailing (`kaskada_on=0`, `hyp_Eb=0`), one hyperon channel at a time, on the configuration of the signal production with the flux weight; GENIE is taken from the generator files of the QE production with the relevant hyperon kept (`genie_to_hepmc.C`). In-FV $\bar{\nu}_\mu$ CC events with a primary hyperon per data POT (`GenRate.C`):

	NuWro	GENIE	NuWro/GENIE
Λ	223.7 ± 2.9	209.5 ± 1.4	1.07
Σ^0	47.9 ± 0.6	81.7 ± 0.6	0.59
Σ^-	115 ± 2	194.9 ± 1.4	0.59

Σ^-/Σ^0 is 2.40 in NuWro and 2.38 in GENIE: the isospin factor 2 times argon's N/Z (2.44). The NuWro/GENIE ratio rises the same way with E_ν for Λ and Σ^0; only the normalisation differs, so neither the flux nor the nucleus is the cause.

Same couplings. Both generators use $f_1^\Lambda = -\sqrt{3/2}\,F_1^p$, $g_1^\Lambda = -(1+2x)g_A/\sqrt{6}$, $f_1^{\Sigma^0} = -(F_1^p + 2F_1^n)/\sqrt{2}$, $g_1^{\Sigma^0} = (1-2x)g_A/\sqrt{2}$, $\Sigma^- \sqrt{2} \times \Sigma^0$, $x = F/(F+D) = 0.365$ and $M_A = 0.99$–1.0 GeV (GENIE `LwlynSmithFF::Strange*`, NuWro `ff.cc`). NuWro contracts the full tensors (`Singh_Model`); GENIE uses a closed form.

Independent calculation. `HypXsecWeights.py` evaluates $L_{\mu\nu}H^{\mu\nu}$ from explicit Dirac matrices, L from $\bar{v}(k_1)\gamma^\mu(1-\gamma_5)v(k_2)$ and H for $F_1\gamma^\mu + i\xi F_2\sigma^{\mu\nu}q_\nu/2M + F_A\gamma^\mu\gamma_5$. On CCQE it gives 1.06 (ν) and 0.41 ($\bar{\nu}_\mu$) $\times 10^{-38}$ cm^2 at 1 GeV, the Llewellyn Smith values. Free proton, 10^{-40} cm^2:

	trace			GENIE spline		
E_ν	Λ	Σ^0	Σ^0/Λ	Λ	Σ^0	Σ^0/Λ
1 GeV	1.33	0.25	0.19	1.02	0.46	0.45
2 GeV	2.15	0.41	0.19	1.71	0.76	0.44

Where GENIE goes wrong. A transcription of `PaisQELLambdaPXSec::XSec` (identical in v3_00_04_ub4 and v3_00_06) reproduces GENIE's Λ spline (1.05 against 1.02 at 1 GeV; its Σ^0 is 15% under GENIE's, which has $G_E^n \neq 0$). With the final-state mass set to the nucleon's it agrees with the trace to 0.5%. As the mass is raised to the hyperon's, the trace cross section falls, as phase space requires, while the Pais Σ^0 cross section rises (0.49 → 0.60 at 2 GeV, against 0.50 → 0.41): the error is in the $M_Y - M_N$ terms.

Polarisation. The spin projector $(1 + \gamma_5 \not{s})/2$ on the hyperon, in its rest frame reached from the struck-nucleon frame, gives P_L, P_T (in the production plane, towards the $\bar{\nu}_\mu$) and $P_N = 0$. The conventions are checked against explicit helicity spinors and on a point-like $V-A$ current ($|\vec{P}| = 1$ at every angle, $P_L \to -0.90$ for a fast hyperon; helicity -1 is forbidden only in the exactly backward corner). The M1 decay passes the polarisation on as $\vec{P}_\Lambda = -(\vec{P}_\Sigma \cdot \hat{n})\hat{n}$. The rebuilt decay directions give 75.4 ± 10 MeV/c for the Λ in the Σ^0 frame (74.5 expected) and 108 ± 26 MeV/c for the nucleon in the Λ frame (100.6).

NuWro's cascade on Λ. Of primary Λ, 64% leave the nucleus as Λ, 32% are jailed, 4% convert to Σ; of primary Σ^0 24% survive.

A.1 The fixed GENIE

`genie::HypQELTracePXSec` (directory `genie_gen/hypfix` of the production area) replaces `PaisQELLambdaPXSec` for the `QEL-CC-LAMBDA` generator: same channels, form factors, Cabibbo angle, integrator and nuclear scaling, with $L_{\mu\nu}H^{\mu\nu}$ evaluated from Dirac matrices. GENIE is not rebuilt: the class lives in its own library, autoloaded by ROOT through its rootmap, and XML overrides on `GXMLPATH` (`ModelConfiguration.xml`, `master_config.xml`) point the generator at it. 387 new splines cover the 44 nuclei of the production spline file for $\bar{\nu}_e$, $\bar{\nu}_\mu$, $\bar{\nu}_\tau$ (250 knots to 1000 GeV), merged into a copy of `gxspl-FNALsmall.xml`. Validation:

- the C++ contraction reproduces the trace exactly (1 GeV: Λ 1.330, Σ^0 0.252, Σ^- 0.499 $\times 10^{-40}$ cm^2);
- GENIE splines, new/Pais: Λ 1.268/1.218/1.189 at 1/2/5 GeV against 1.269/1.216/1.185 expected; Σ^0, Σ^- 0.665–0.698 against 0.656–0.687 ($G_E^n \neq 0$ adds $\sim 2\%$);
- a 360-event FHC `lar` generation: primary Λ $59.2 \pm 2.6\%$, Σ^- $30.1 \pm 2.4\%$, Σ^0 $10.7 \pm 1.6\%$, against 42.8/40.4/16.7% with Pais and about 58/30/12% expected. Its POT per event is 1.8σ below the +9% expected from the lower total cross section; to be checked on the production.

At the same chunk count the regenerated samples hold about 0.7× the Σ^0 statistics of the old ones.

B Signal-definition thresholds

`ThreshStudy.C` takes each Σ^0 event of the signal productions with $\Lambda \to p\pi^-$ (true vertex in the FV, muon above threshold, no other meson or strange particle) and records the true proton, π^- and photon momenta and whether a PFP truth-matched to each (by `pfp_true_pdg` and `pfp_true_origin`: Λ daughter, Σ^0 daughter) was reconstructed, and as the right type (track score above 0.5 for p and π^-, below for the photon). GENIE, 51 096 events:

	proton				π^-				photon	
p [GeV/c]	found	track	sel.	p [GeV/c]	found	track	sel.	E [GeV]	found	shower
0.20–0.25	8.6	5.9	0.00	0.06–0.08	5.9	4.4	0.21	0.02–0.04	37	30
0.25–0.30	21	17	0.08	0.08–0.10	28	22	0.55	0.04–0.06	50	39
0.30–0.35	38	32	0.17	0.10–0.12	50	38	1.11	0.06–0.08	57	47
0.35–0.40	50	42	0.69	0.12–0.14	59	47	1.49	0.08–0.10	63	52
0.40–0.45	58	49	1.04	0.14–0.16	64	52	2.24	0.10–0.12	67	58
0.50–0.60	68	59	2.41	0.16–0.18	66	53	2.98	0.12–0.15	70	61
0.60–0.70	73	66	3.86	0.18–0.20	64	50	3.34	0.15–0.20	69	62

(Percentages; "sel." is the fraction of all such events passing the Λ-style selection.) The threshold is the momentum at which the right-type efficiency reaches half its plateau: proton 0.340 GeV/c (plateau 67%; KE 60 MeV), π^- 0.095 GeV/c (52%; KE 30 MeV), photon 0.030 GeV (59%). NuWro gives the same three values to the bin. The proton and pion thresholds each remove 24–26% of the $\Lambda \to p\pi^-$ signal but only 0.5–1.3% (proton) and 3.7–4.3% (pion) of the selected signal; the photon threshold removes nothing measurable, since the Σ^0 photon is 50–131 MeV (5–95%). The plateaus are below 100% because of overlapping tracks near the vertex, pion interactions and track/shower misclassification, which are selection efficiency, not threshold. The vetoes (any meson, any strange particle besides the Σ^0) remove $\leq 0.01\%$ of either signal sample.

C Samples: history and validation

NuWro run3b production. 2500 jobs of 100 NuWro Σ^0 events overlaid on run3b `ext_unbiased` data through gen→reco2, with hypana and PeLEE ntuples; it replaced the Run 1-EXT sample in October 2026 (3.3× the raw statistics, cosmics from a period the background overlay covers). POT agrees between the two formats to four digits; no EXT event is used twice.

Background sample corrections. The file first used as "Run 1 dirt" was FHC dirt (its E_ν spectra match FHC, $\chi^2/\mathrm{ndf} \approx 1$, not RHC, 90–660); it was replaced by RHC dirt, which is on the old flux (`ppfx_cv` profile) and takes the NuWro map. The Run 3 overlay (55% of the statistics) was missing. EXT is scaled by beam-on triggers (23 534 937 per 1.10823×10^{21} POT); data files have empty `backtracked_*` vectors, and 1 655 repeated events are dropped. The overlay's own 1 038 raw in-FV GENIE Σ^0 events are kept out of B and counted as the GENIE signal prediction.

GENIE Σ^0 QE sample. GENIE `LambdaCCQE`, tune G18_10a_02_11a, official dk2nu RHC flux, primary Σ^0 selected at generation, same overlay chain; 2506 jobs, 1.23005×10^{24} POT. Before the cross-section correction its in-FV rate, 81.85 ± 0.28, agreed with the overlay's own Σ^0 (83.39 ± 3.69, 0.98 ± 0.04) and in shape (Fig. 6). The overlay's non-QE Σ^0 (0.8%) come with a kaon, which the signal definition vetoes, so the QE sample is the complete GENIE prediction. The previous GENIE hyperon sample (`sig_genie.list`) held 15 341 events but only 754 distinct interactions, each overlaid about 20 times, at 1.32× the overlay's rate; everything that used it was redone.

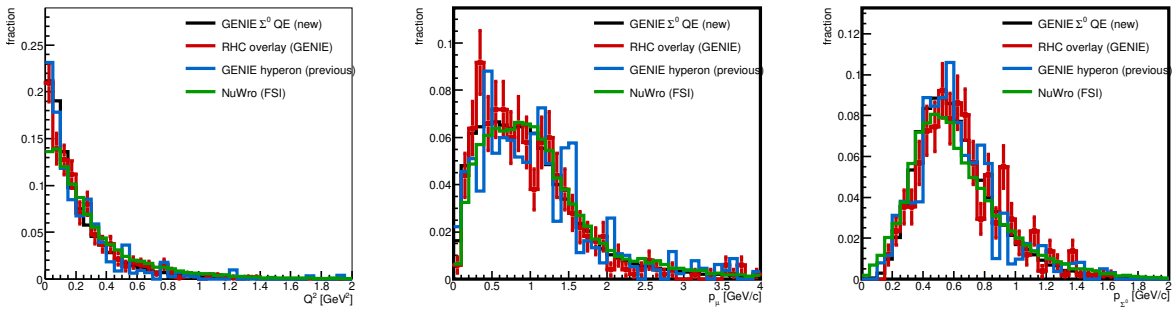


Figure 6: Truth Q^2, muon momentum and Σ^0 momentum of in-FV $\bar{\nu}_\mu$ CC Σ^0, area-normalised: the GENIE Σ^0 QE sample, the plain RHC overlay, the previous GENIE hyperon sample and NuWro (before the cross-section correction).

The rate comparison before the correction. Before GENIE's cross section was corrected, the 6.5× gap between the GENIE overlay (83.3) and NuWro (12.8) was decomposed as FSI (4.19×), flux weight (1.48×) and production (1.05×). A NuWro control run with `dyn_hyp_cc=0` `dyn_qel_cc=1` gives 7.539×10^{15} POT per CCQE event against the overlay's 8.835×10^{15} on the same footing, so NuWro's generic rate is within 17% of GENIE's. With the correction the gap is 4.4× and production agrees to 0.96.

D PeLEE BDT development

Out of sample on NuWro (signal statistics limit the BDT, so every gain came from more or better-used events):

training	AUC	B/B_0 at signal efficiency			peak $S/\sqrt{S+B}$	
		50%	40%	20%	value	cut
no EXT, half fold	0.8894	0.0412	0.0253	0.0067	0.0993	0.690
EXT, half fold (3 seeds)	0.8956–0.8966	0.0378–0.0385	0.0228–0.0236	0.0059–0.0062	0.1037–0.1049	
EXT, 95% of fold	0.9000	0.0358	0.0215	0.0061	0.1077	0.759
clamped inputs	0.9057	0.0336	0.0203	0.0063	0.1102	
clamped + GENIE signal (adopted)	**0.9086**	**0.0317**	**0.0186**	**0.0055**	**0.1146**	**0.779**
clamped + GENIE, seed 1	0.9093	0.0312	0.0186	0.0055	0.1147	

- **EXT in training**: +5% figure of merit, almost all EXT rejection.
- 95% **of each fold**: TMVA's default trains each fold on half its events (25% of all); +4%, every full-fold seed beats every half-fold seed.
- **Input clamp**: one ±FLT_MAX value leaves a variable unusable in a node because TMVA's cut grid is evenly spaced; with bagging almost every root node contained one of five events with infinite masses. The five pair distances carry a 10^5 sentinel in 21–59% of events, `ShrSubClusters` INT_MAX in 29%.
- **GENIE signal in training**: more distinct signal helps even judged on NuWro; −5% background at 40–50% efficiency.

The BDT trains on unweighted GENIE events; with the hyperon weights its scores are unchanged and its yields correctly weighted, so it was not retrained.

E What limits the PeLEE purity

Measured on earlier versions of the chain; the fractions carry over.

- **The surviving background is π^0 events**: resonant 52%, DIS 24%, QE 14%, MEC 7%, cosmic 3%. A π^0 supplies the fake photon and its hadrons the fake proton and pion.
- **The candidate assignment is the weak link**: in background the muon candidate is right 81% of the time, the proton 58%, the pion 31%, the photon 29%.
- **The signal photon is soft**: $E_\gamma = (m^2_{\Sigma^0} - m^2_\Lambda)/2m_{\Sigma^0} = 74.5$ MeV, against 140 MeV for background photons. Soft-photon cuts help at preselection but not on top of the BDT, which already uses the energy; a π^0 mass veto rejects only 9% because it needs two showers; the Molière angle adds 1%.
- **Methods from the single-photon analyses**: the second-shower-veto variables are BDT inputs; one BDT per background class and a `HasPair` split were tried and do not help; the conversion distance is the Λ-style photon gap; the π^0 normalisation (GENIE over-predicts NCπ^0 by 25%) belongs in the covariance and a sideband constraint.

F Reconstruction study on HyperonProduction ntuples

The PeLEE truth record never contains the $\Sigma^0 \to \Lambda\gamma \to p\pi^-\gamma$ chain (zero of 17 721 signal events have a backtracked π^-). Hypana writes the full chain and per-PFP truth; `HypInput.h` maps it onto the PeLEE names. Three format traps: no implicit neutrino-slice requirement (an explicit `flash_match_nu_slice_ID` ≥ 0 stage brings the chains to 6–8% agreement), `trk_energy_proton` is the proton momentum (with the conversion the Λ mass width falls from 23.5 to 7.8 MeV), and `pfp_true_origin` is the analyser's own enum (1 neutrino primary, 2 Λ daughter, 3 other, 5 Σ^0 daughter, −1 unmatched).

With both the reconstructed and true vertex in the FV and $\Lambda \to p\pi^-$ in truth (3704 signal events), the four particles are reconstructed as: μ^+ 91.8%, p 60.9%, π^- 47.3%, γ 62.4%; all four together 18.1%, the ceiling for any selection requiring all four. Track/shower: the pion is a track 54% of the time, and the photon a track 15%, which feeds the photon contamination of the pion slot. The Λ flight from the primary vertex is not measurable (median true flight 1.85 cm against 5.5 cm vertex resolution, $r = -0.014$), which is why the Λ-style chain fits the decay vertex from the two tracks instead.

G Λ-style chain: development and checks

Island connectivity. The original 3×40 bridged the decay V to the muon for flights below ~ 3 cm:

true flight	1–2 cm	2–3 cm	3–5 cm	5–10 cm
3×40	19%	45%	70%	86%
2×20	43%	80%	85%	91%

On one plane the figure of merit is flat in the connectivity; the gain is from two planes (6.8× background rejection against 3.7×). Tuned on the Run 3 list and scored on the independent extension, 2×20 is 0.004 from the top of a plateau. The two-plane pass rate is 50% for both generators.

Out-of-sample checks. Without the photon gap, the Run 3 list gives $B = 4.15 \pm 1.31$ and the extension 7.6 ± 1.4; every earlier stage agrees per POT to 1–4%, so the low Run 3 value was a fluctuation. The gap halves the extension's background at 96% signal efficiency. A Michel tag removes 87% of true μ^- but costs 22% of signal.

Sidebands (one-plane point): mass sideband +40 to +200 MeV $S/B = 0.2\%$, α sideband 0.9%, island-fail region 1.1%, all RES/DIS-dominated like the signal region; the mass sideband exists only above the peak ($m(p\pi) \geq 1.078$ GeV).

EXT factorisation.

step	EXT raw	EXT P	MC cosmic raw	MC cosmic P	MC all-bkg P
MuonCandidate	76 ($11\,419 \pm 1\,310$)	–	4 220	–	–
pair	8	0.105 ± 0.035	571	0.135 ± 0.005	0.42
Λ mass	3	0.38 ± 0.17	165	0.289 ± 0.019	0.32
pointing	0	0 of 3	7	0.042 ± 0.016	0.043
island + gap	0	–	0 of 7	–	0.0326

Hypana EXT at MuonCandidate is 1.7× the PeLEE beam-off there, because hypana's `topological_score` is the highest of any slice; the MC cosmics use the same definition.

Λ cross-check. Reimplementing the Λ CCQE chain on our files (`LambdaSel.C`) reproduced its efficiency to 0.79×, but the GENIE signal normalisation came out 1.61× the published one, on the duplicated hyperon sample. It has to be redone on the QE sample with the corrected cross section, which moves GENIE's Λ by ×1.26.

Muon-track photon veto. With the attached zone starting at the vertex (−5 cm along the muon), 23% of signal photons count as attached, against 50% of real-Λ and 33% of other background photons; re-picking the photon among the non-attached showers costs 18% of signal. Starting the zone 10, 20 or 30 cm along the muon flags 8.5, 7.5, 7.2% of signal, 4, 3, 3 of 10 real-Λ events and 9.3, 8.5, 8.0% of other background at the photon-gap stage. On top of the PID cut the 20 cm veto gives $S = 0.484$, $B = 1.99$: median limit 9.03 (the median n stays 2 while S drops), mean 9.32 against 9.38.

The shower requirement. The preselection requires $0 < n_{\mathrm{shr}} < 4$ and $n_{\mathrm{trk}} < 5$ (`PreselFlow.C`). Of the in-FV GENIE signal (51.1 events) 84% has a flash-matched neutrino slice, 82% a reconstructed vertex in the FV, 75% passes the topological score, 57% the shower/track requirement and 50% has a muon candidate; the shower requirement removes 7.8 events, almost all with no shower at all. Relaxing the shower maximum, the track maximum or the topological score is worth 2–5% each. The events with no shower (`IslandProcess.C showerMode=1`) as a separate bin, at the PID operating point:

stage	S GENIE	S NuWro	raw bkg	B_{MC}
no shower, $n_{\mathrm{trk}} < 5$	10.70	2.71	213 731	34 949
muon candidate	9.38	2.40	174 614	28 538
pair	3.31	0.867	51 839	8 472
Λ mass	1.84	0.479	17 754	2 902
pointing	0.499	0.125	606	99.9
island ≥ 2 planes	0.244	0.065	25	4.73
proton PID	0.213	0.059	21	4.04

EXT, factorised as for the main bin from the hypana EXT run in the same mode (161 muon candidates against 76 with a shower), is 1.53 ± 0.88, so $B = 5.64$. The bin's background passes the PID cut 84% of the time (main bin 52%): it is mostly genuine $p\pi^-$ pairs, and there is no photon gap to apply. Pooled with the main bin the mean expected limit worsens (9.98 against 9.39 GENIE); as a second bin it would improve by 7% (GENIE) and 8% (NuWro). Not used.

Background photons. The chain's photon in the real-Λ background is never a Σ^0 photon. Of 10 real-Λ events at the photon-gap stage, half have $e^{\pm}$ (Michel positrons and muon δ rays), 30% another photon, 10% a cosmic. In signal the chain picks the true Σ^0 photon 80% of the time (median 115 hits on the collection plane, 54 MeV, 24 cm long); the real-Λ "photons" are similar (75 hits, 29 MeV, 14 cm) and the rest of the background larger (133 hits, 87 MeV, 28 cm).

G.1 The last three cuts and the shower ceiling

`LastCutsScan.py` varies the photon gap, the island planes, the proton `llrP` and the shower ceiling with the earlier cuts at the operating point, the signal of Sec. 2.1, every weight, the out-of-definition Σ^0 of each model in its background, and EXT and NuWro non-signal hyperons scaled with the overlay MC. It reproduces the operating point of Table 6. Table 9 varies one cut at a time; "mean" is the expected limit averaged over $n \sim \mathrm{Pois}(B)$, steadier than the median at $B \approx 2$–3.

Why up to three showers. The ceiling $0 < n_{\mathrm{sh}} < 4$ was not chosen for this chain: it comes from the original cut-based selection (loosened there from < 3), through the PeLEE chain. A truth match of every shower-like PFP (`ShowerOrigin.C`) shows the extra showers in signal are the signal event's own activity. Of the defined GENIE signal at the muon stage 61% have one shower, 31% two and 8% three; their extra showers are secondaries (38%: fragments of the photon shower and the Michel e^+ of the μ^+), the Λ pion (25%) or proton (16%) reconstructed as a shower, cosmics (9%), a split piece of the Σ^0 photon (7%) and the muon (5%). The events whose Λ daughter became a shower fail the pair requirement later; at the final selection 75% have one shower, 22%

Table 9: Λ-style chain, one cut varied at a time from the operating point (bold), $f = 15\%$, $g = 20\%$. B in the NuWro model; limits GENIE / NuWro.

cut	value	S_G	B	raw bkg	median μ_{UL}	mean μ_{UL}
photon gap	none	0.497	4.59	23	10.46 / 42.9	12.14 / 49.5
	> 1 cm	0.486	3.25	16	9.95 / 40.4	10.95 / 44.2
	> 1.5 cm	0.483	3.06	15	10.18 / 41.2	10.77 / 43.3
	$>$ **2** cm	**0.475**	**2.49**	**12**	**8.73 / 35.2**	**10.22 / 40.8**
	> 3 cm	0.463	2.29	11	9.14 / 37.5	10.22 / 41.7
	> 4 cm	0.444	2.10	10	9.72 / 40.3	10.38 / 42.7
	> 6 cm	0.404	1.72	8	11.11 / 45.4	10.71 / 43.4
island	≥ 1 plane	0.702	11.12	56	11.81 / 49.3	12.55 / 52.2
	≥ 3 planes	0.209	0.86	4	17.62 / 69.3	17.19 / 67.4
	all working planes	0.221	1.25	6	15.99 / 62.7	17.81 / 69.6
`llrP`	none	0.514	4.59	23	10.12 / 39.8	11.75 / 46.0
	< 0.5	0.498	3.82	19	11.16 / 44.0	11.32 / 44.3
	< -0.25	0.426	2.05	10	10.16 / 41.7	10.71 / 43.7
	< -0.5	0.339	1.86	9	13.03 / 52.1	13.04 / 51.8
showers	$n_{sh} \leq 2$	0.459	2.29	11	9.19 / 36.7	10.30 / 40.8
	$n_{sh} = 1$	0.356	1.06	5	10.12 / 40.8	10.65 / 42.6

two and 3% three, the extra ones now 64% secondaries, 13% cosmics, 8% photon splits. NuWro gives the same. Requiring exactly one shower halves the background but loses a quarter of the signal and worsens the limit by 4% (mean); $n_{sh} \leq 2$ changes nothing measurable. The ceiling is kept; $n_{sh} \geq 4$ is 0.7% of the signal at preselection.

H The hypana BDT chain

A BDT on the hypana candidates (53 inputs, including the island variables), used for the truth studies of App. F, not for the limit. Trained on the GENIE QE signal with two-fold parity cross-validation, AUC 0.855. With the hyperon weights (matched by event order to the island caches, which carry event identifiers) at its cut 0.554: $S = 11.32$ (GENIE, no PPFX) on $B = 737.4$, purity 1.5%.

I Methods kept for later

A global hypothesis fit (not built): every PFP eligible for every role, all assignments scored with a combined $-2 \ln L$ from the Λ and Σ^0 mass constraints, per-role PID templates, the p–π DCA and priors; perfect assignment given what is reconstructed is worth 1.54× on events with four PFPs. Its central idea, joint assignment, is realised by the pair scoring of both chains. The measured ingredients are in `HypFitParams.h`; the masses must be kept out of any BDT and the fit run on background alone to check for an induced peak.

Fake data by Poisson throw. The PeLEE "data" slot holds the GENIE QE sample scaled by POT: a generator test of the efficiency, not a closure test. The CC1μ1π Poisson-throw scheme (`throw_perrun_w.C`: $n \sim \text{Pois}(w_{cv}\alpha)$ per MC event, EXT at its scale) applied to all samples gives the statistical closure test; injecting the GENIE signal into a NuWro-based throw gives the model test.

Framework port. A `SelectionBase` subclass in `xsec_analyzer` was planned for its systematics machinery; it is not started and no longer on the critical path, since the PeLEE ntuples carry the

variable	μ^+	p	π^-	γ
`trk_shr_score`	1.00	0.90	0.80	**0.11**
`trk_llrpid`	**0.96**	−0.29	0.14	0.56
`trk_pid_chipr`	180	**16.4**	96	192
`three_plane_dedx`	2.47	**10.3**	7.52	3.30
`rms_deflection`	0.01	0.01	0.01	**0.08**
`moliere_avg`	1.78	6.18	8.95	**13.3**
`trk_length`	**139**	12.8	10.8	14.2

Table 10: Per-role PID templates, medians on truth-matched PFPs. The π^- has no distinctive signature.

multisim weights and `Sensitivity.C` provides the limit.

J Change log

2026-10-15 NuWro run3b signal production replaces the Run 1-EXT sample.

2026-10-17 RHC dirt replaces FHC dirt; Run 3 overlay and EXT added; overlay signal removed from B.

2026-10-18 PeLEE BDT retrained with EXT and on 95% of each fold.

2026-10-19 Detector variations; Λ-style chain and its EXT estimate; expected sensitivity.

2026-10-30 GENIE Σ^0 QE sample replaces the duplicated hyperon sample; PeLEE BDT retrained with clamped inputs and GENIE signal (cut 0.759 → 0.779). GENIE's hyperon cross section found wrong; cross-section and polarisation weights applied everywhere (GENIE signal 13.47 → 8.86, limits 1.5× weaker).

2026-10-31 GENIE fixed (`HypQELTracePXSec`); RHC and FHC GENIE signal regeneration started. Λ-style background weighted; background composition; FHC signal estimate; note restructured around the status (the dated version is archived in `report/archive`).

2026-11-01 Λ-style background re-analysed by event truth: real Λ 46% (QE 33%, associated production 13%; previously counted as Δ), plan rewritten.

2026-11-01 Λ-style last three cuts and shower ceiling re-scanned with the full definition: operating point kept (on a plateau; App. G.1).

2026-11-01 One signal definition for both chains: $\Lambda \to p\pi^-$, $\mathrm{KE}_p > 60$, $\mathrm{KE}_\pi > 30$, $E_\gamma > 30$ MeV (measured turn-ons), no other meson or strange particle; out-of-definition Σ^0 as background per model. PeLEE BDT retrained for it (cut 0.830). The Λ-style chain gives the better limit at every f.

2026-11-01 Λ-style proton PID cut adopted (B 4.56 → 2.48, limit 9.3 → 8.1× GENIE at $f = 15\%$); the PeLEE/Λ-style crossover moves to $f \approx 21\%$. Muon-track photon veto, loosened cuts and a no-shower second bin measured and not used.